RAG-001: Retrieval and hallucination

Retrieval-augmented generation reduces hallucination by retrieving relevant passages before an answer is generated. The retrieved evidence constrains the answer to what the source supports instead of encouraging unsupported claims.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

RAG-002: Dense similarity retrieval

Dense vector representations support similarity search over an embedding index. Similarity ranking selects passages whose meaning is close to the question before generation begins.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

RAG-003: Grounding in external knowledge

Grounding an answer in external knowledge improves factual accuracy because generation can rely on retrieved evidence rather than model parameters alone. External knowledge makes factual statements traceable to a source.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

TF-001: Parallel self attention

Self attention compares token positions together, so a transformer can process a sequence in parallel. This removes the sequential dependency imposed by recurrent processing.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

TF-002: Transformer versus recurrent computation

The transformer architecture replaces recurrent computation with attention over the whole sequence. A recurrent network processes positions in order, while a transformer can attend across positions directly.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

TF-003: Query, key, and value projections

Each token is projected into query, key, and value vectors. A query compares with keys to find relevant positions, and the resulting weights combine the corresponding value vectors.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

CMP-001: Training efficiency

A transformer avoids the recurrent dependency that limits parallel hardware use. Compared with recurrent neural networks, this enables faster training when many sequence positions are processed concurrently.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

CMP-002: Retrieval and language models

A retrieval-based language model consults an external corpus at inference time. A purely parametric language model instead answers from weights learned during training without retrieval from that corpus.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

KG-001: Knowledge graph support

A knowledge graph connects related entities and concepts. In a retrieval pipeline, graph traversal can surface related evidence that a dense or lexical match did not retrieve.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.

CHK-001: Chunking documents

A document is split into small passages before embedding. Focused passages let retrieval return the specific evidence needed for a question instead of an entire document.

This page intentionally contains the evaluation evidence for one curated golden-set case. It is synthetic test content, not a research publication.